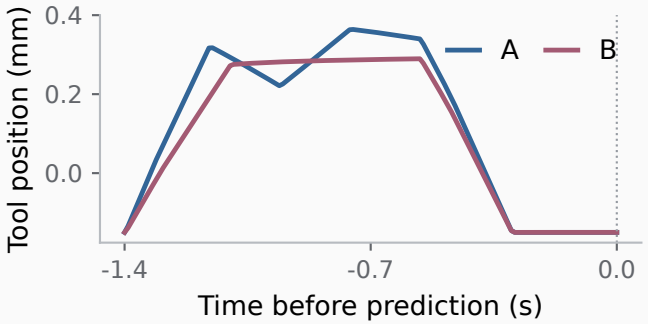
2D contact experiment

Load → Hold → Release → Wait → Recontact

Loading histories

Different loads, matched endpoints



Match position, velocity, force and both lateral markers.

Lateral residual: 2.18 µm (limit 5 µm)

State estimation

Current measurements

0.2 s observation window

Full loading history

True internal state

Simulation reference

Repeat over window lengths, noise levels and loading excitation.

Future response

Slow press

Fast press

Hold

Release

Compare predicted force and lateral displacement with the continued simulation.

All estimates use the same model for future prediction. Full replay knows the relaxed start.

Synthetic material: relaxation times 0.12 s and 0.90 s. Eight test histories.